

Final Year Project Proposal

Title: AI-Based Intrusion Detection System with Real-Time SOC Monitoring Dashboard

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Abstract

With the rapid growth of digital systems and internet connectivity, cyber threats have also increased significantly. Traditional security mechanisms such as firewalls are no longer sufficient to detect modern and sophisticated attacks. Intrusion Detection Systems (IDS) play a vital role in identifying malicious activities in networks or systems. This project proposes the development of an AI-Based Intrusion Detection System integrated with a Real-Time SOC (Security Operations Center) Dashboard. The system will utilize machine learning algorithms to analyze system or network logs, detect abnormal behavior, and classify activities as normal or malicious. It will also provide real-time alerts, visual analytics, and historical attack monitoring. The goal of this project is to develop an intelligent, scalable, and educational cybersecurity monitoring system that enhances threat detection capabilities using artificial intelligence.

1. Introduction

Cybersecurity has become a critical concern in modern computing environments due to the increasing number of cyberattacks such as brute force attacks, denial of service (DoS), unauthorized access attempts, and malware infections. Organizations require intelligent systems that can continuously monitor their infrastructure and detect suspicious activities in real time.

An Intrusion Detection System (IDS) is designed to monitor network or system activities and identify potential security breaches. However, traditional IDS solutions are often based on static rules and signature-based detection, which limits their ability to identify unknown attacks. To overcome this limitation, Artificial Intelligence (AI) and Machine Learning (ML) techniques can be integrated into IDS to improve detection accuracy and adaptability.

This project focuses on building an intelligent IDS using AI techniques combined with a real-time dashboard for visualization and monitoring.

2. Problem Statement

Existing security systems mostly rely on predefined rules and signatures to detect threats, which makes them ineffective against new or unknown attack patterns. Additionally, many commercial IDS solutions are complex, expensive, and not suitable for academic learning or small-scale environments. Therefore, there is a need for a lightweight, intelligent, and cost-effective intrusion detection system that can learn from data and detect anomalies in real time.

3. Objectives

The main objectives of this project are:

- To design and develop an AI-based Intrusion Detection System
 - To collect and analyze system or network logs for anomaly detection
 - To train machine learning models for classifying normal and malicious behavior
 - To detect common cyber threats such as brute force attacks, DoS attacks, and suspicious login attempts
 - To provide real-time alert generation for detected threats
 - To visualize security events using an interactive SOC dashboard
 - To maintain historical logs and attack trends for analysis
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4. Scope of the Project

The proposed system focuses on monitoring and analyzing system or network behavior to detect malicious activities using AI techniques.

Limitations:

- The system will focus on known and pattern-based attacks using available datasets
 - It will not fully replace enterprise-level security systems
 - It is designed primarily for educational and small-scale environments
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5. Methodology

The system will be developed using the following phases:

Phase 1: Data Collection

- Collect network/system log datasets such as CICIDS2017 or KDD Cup 99
- Extract relevant features such as login attempts, IP address behavior, request frequency, and time intervals

Phase 2: Data Preprocessing

- Clean and normalize data
- Handle missing values
- Select important features for model training

Phase 3: Model Training

- Train machine learning models such as Random Forest, Decision Tree, or Logistic Regression
- Evaluate model accuracy using test datasets

Phase 4: Detection Engine

- Deploy trained model for real-time or simulated input data
- Classify activity as normal or malicious

Phase 5: Alert System

- Generate alerts when suspicious activity is detected
- Store logs for future analysis

Phase 6: SOC Dashboard

- Visualize alerts, attack types, and system status using graphs and charts
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6. Tools and Technologies

Programming Language:

- Python

Machine Learning Libraries:

- Scikit-learn
- Pandas
- NumPy

Backend Framework:

- Flask / Django

Visualization Tools:

- Chart.js
- Plotly

Dataset Sources:

- CICIDS2017
 - KDD Cup 99
 - UNSW-NB15
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7. System Architecture

Data Source → Preprocessing → Machine Learning Model → Detection Engine → Alert System → SOC Dashboard

8. Expected Outcomes

- AI-based intrusion detection system
 - Real-time or simulated attack detection
 - Interactive SOC dashboard
 - Alert generation system
 - Historical attack analysis and reporting
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9. Novelty and Innovation

- Integration of AI-based detection with SOC-style dashboard
 - Ability to detect anomalies instead of only rule-based attacks
 - Real-time visualization of cyber threats
 - Lightweight and educational alternative to enterprise IDS systems
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10. Future Enhancements

- Deep learning-based attack detection
 - Cloud-based deployment for multi-user access
 - Automated response system (future IPS integration)
 - Advanced behavior analytics
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11. Conclusion

The proposed AI-Based Intrusion Detection System provides an intelligent and efficient approach to cybersecurity monitoring. By integrating machine learning techniques with real-time visualization, the system enhances the ability to detect and analyze cyber threats effectively. This project serves as both an educational tool and a practical implementation of modern cybersecurity concepts.

12. References

- CICIDS2017 Dataset Documentation
- KDD Cup 99 Dataset
- Scikit-learn Official Documentation
- IEEE Cybersecurity Research Papers
- NIST Cybersecurity Framework